

CONTAINERIZING A LIBRARY APP USING PYTHON CODE

- Launch an EC2 instance
 - ami : default
 - instance type : 2 CPU's & 4 GB RAM
 - EBS : 18 GB SSD
- Install docker
 - `yum install docker -y`
 - `systemctl start docker`
- Install git and clone repo
 - `yum install git -y`
 - `git clone https://github.com/shaikmustafa77/python-library-app.git`
 - `cd python-library-app`
 - `git checkout master`
 - `ll` (to list the code)
- Build the Images:
 - application image : `docker build -t appimage .`
 - auth image : `docker build -t authimage auth/`
 - book image : `docker build -t bookimage book/`
 - borrow image : `docker build -t borrowimage borrow/`
 - db image : `docker build -t dbimage database/`
- Create Network:
 - `docker network create mynet`
- Create containers for our application in mynet network:
 - db container : `docker run -d --name db --network mynet -p 3306:3306 dbimage`
 - Auth container : `docker run -itd --name auth_service --network mynet -p 5001:5001 authimage`
 - Book container : `docker run -itd --name book_service --network mynet -p 5002:5002 bookimage`
 - Borrow Container : `docker run -itd --name borrow_service --network mynet -p 5003:5003 borrowimage`
 - APP Container : `docker run -itd --name frontend --network mynet -p 5000:5000 appimage`
- Access the application by using public-ip:5000

composefile

compose file:

version: "3"

services:

db:

container_name: db

build: ./database/

ports:

- "3306:3306"

networks:

- mynet

auth:

container_name: auth_service

build: ./auth/

ports:

- "5001:5001"

networks:

- mynet

books:

container_name: book_service

build: ./book/

ports:

- "5002:5002"

networks:

- mynet

borrow:

container_name: borrow_service

build: ./borrow/

ports:

- "5003:5003"

networks:

- mynet

frontend:

container_name: frontend

build: ./

ports:

- "5000:5000"

networks:

- mynet

networks:

mynet:

driver: bridge